

Factoring Higher Order Polynomials

Factoring higher-order polynomials in $\mathbb{Z}$ (the set of rational numbers) can be simplified using two powerful tools: the **Rational Roots Theorem** and **Synthetic Division**. Here's an overview of these techniques and how they work together.

The **Rational Roots Theorem** provides a method to identify possible rational roots of a polynomial equation with integer coefficients.

Rational Roots Theorem. If $\frac{p}{q}$ is a rational root of

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

where $a_i \in \mathbb{Z}$ for $0 \leq i \leq n$, then p is a factor of the constant term a_0 and q is a factor of the leading coefficient a_n .

Thus, all potential rational roots can be listed as:

$$\pm \frac{\text{Factors of } a_0}{\text{Factors of } a_n}.$$

Example: For $P(x) = 2x^3 - 3x^2 - 5x + 6$, the constant term is 6, and the leading coefficient is 2. Potential rational roots are:

$$\pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{1}{2}, \pm \frac{3}{2}.$$

Once you have a list of possible rational roots, **Synthetic Division** helps determine which ones are actual roots. Synthetic division is a streamlined process for dividing a polynomial $P(x)$ by a linear factor $x - c$. If the remainder is zero, c is a root of the polynomial, and $x - c$ is a factor.

Steps for Synthetic Division:

1. Write the coefficients of $P(x)$ in descending order of degree.
2. Place the candidate root c to the left of a division box.
3. Bring down the first coefficient.
4. Multiply c by the value just written, and write the result under the next coefficient.
5. Add the column and repeat until all coefficients are processed.
6. The final value is the remainder. If it's 0, c is a root.

Example: Divide $P(x) = 2x^3 - 3x^2 - 5x + 6$ by $x - 2$ (testing if $x = 2$ is a root). Coefficients: 2, -3, -5, 6.

$$\begin{array}{r|rrrr} 2 & 2 & -3 & -5 & 6 \\ & & 4 & 2 & -6 \\ \hline & 2 & 1 & -3 & 0 \end{array}$$

Since the remainder is 0, $x = 2$ is a root, and $P(x) = (x - 2)(2x^2 + x - 3)$.

After finding one root and factoring out the corresponding linear factor, repeat the process:

1. Use synthetic division to reduce the polynomial to a lower degree.
2. Apply the Rational Roots Theorem again to find additional roots.

3. Continue until the polynomial is fully factored or a quadratic factor remains, which can be solved using the quadratic formula.

Example (continued): The quotient from the previous step is $2x^2 + x - 3$. Using the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1^2 - 4(2)(-3)}}{2(2)} = \frac{-1 \pm \sqrt{25}}{4} = \frac{-1 \pm 5}{4}.$$

Roots: $x = 1$ and $x = -\frac{3}{2}$. Final factorization:

$$P(x) = (x - 2)(x - 1)(2x + 3).$$